Monte Carlo Simulation

Objective:
The goal of this project is to implement Monte Carlo simulations to solve real-world problems and explore the concepts of randomness, probability, and statistical analysis through coding and computational techniques. Students will demonstrate their understanding by applying Monte Carlo simulations to different scenarios and evaluating the results.

Instructions:
- Choose one problem from the list below or propose your own topic (must be approved by the instructor).
- Use Python (or another approved programming language) to implement the Monte Carlo simulation.
- Your project should include both code and a report. The report must include the problem statement, methodology, results (including visualizations), analysis, and conclusions.
- You are required to submit both the code and the report.

Problem Set:

1. Estimate the Value of Pi

- Problem: Use Monte Carlo simulation to estimate the value of Pi. You will generate random points inside a square and determine how many fall inside a quarter circle. Use the ratio of points inside the circle to the total number of points to estimate Pi.
- Requirements: Write a function to perform the simulation for different numbers of points (e.g., 10000, 100000, 1000000).
- Visualize the points and the quarter circle on a graph.
- Calculate the accuracy of your estimation compared to the true value of Pi.

2. Stock Price Prediction Using Monte Carlo Simulation

- Problem: Model the future price of a stock using the Monte Carlo method. Simulate the stock price over time using a geometric Brownian motion model. Use historical stock data to estimate volatility and drift.
- Requirements: Retrieve historical stock price data and calculate the drift and volatility.
- Simulate multiple paths for the stock price and visualize them.
- Use the simulation to estimate the future stock price at a specific time (e.g., 30 days from now).
- Discuss the limitations of using Monte Carlo for stock predictions.

3. Risk Analysis for Project Management

- Problem: In project management, Monte Carlo simulations are used to analyze the risk of schedule overruns. You are given the estimated duration for each task in a project and their uncertainties. Use Monte Carlo simulation to estimate the total project duration and the probability of finishing the project on time.
- Requirements: Create a model where each task duration is represented by a probability distribution (e.g., normal distribution with mean and standard deviation).

- Run the simulation for different numbers of trials.
- Visualize the distribution of the total project duration and estimate the probability of meeting the deadline.

4. Simulate Dice Rolls for a Game of Chance

- Problem: Simulate the rolling of two six-sided dice and use Monte Carlo simulation to calculate probabilities of outcomes, such as rolling a sum of 7. Extend this to more complex games, such as craps.
- Requirements: Simulate rolling two dice and compute the probabilities for sums between 2 and 12.
- Extend the simulation to a game like craps and estimate the probability of winning.
- Visualize the distribution of outcomes and discuss how the results compare to theoretical probabilities.

5. Monte Carlo Simulation for Investment Portfolio Optimization

- Problem: Use Monte Carlo simulation to optimize an investment portfolio. Simulate the future performance of a portfolio consisting of multiple assets with different expected returns, volatilities, and correlations. Your goal is to estimate the portfolio's risk and return.
- Requirements: Define a portfolio with different assets (e.g., stocks, bonds) and their historical data (mean returns, volatilities, and correlations). simulate the future returns of the portfolio over a specified time period (e.g., 1 year, 5 years).
- Run multiple simulations to estimate different potential outcomes and create a distribution of portfolio returns.
- Visualize the risk-return trade-off and use the results to suggest an optimal portfolio allocation.

Project Requirements:

1. Code:
  - Must be well-documented with comments explaining each section.
  - Should include functions or classes to organize your work.
  - Use of libraries such as `numpy`, `matplotlib`, or `pandas` (if applicable) is encouraged.

2. Report:
  - Introduction: Briefly explain the problem and why Monte Carlo simulation is an appropriate solution.
  - Methodology: Detail how you approached the problem, including key assumptions, algorithms used, and any relevant formulas.
  - Results: Present your findings, including visualizations such as graphs, histograms, or scatter plots.
  - Discussion: Analyze the results. Discuss the accuracy, efficiency, and any potential improvements.
  - Conclusion: Summarize the key takeaways from your project.

Assessment Criteria:
- Correctness: Does the simulation solve the problem as specified?
- Efficiency: Is the code optimized and efficient for larger datasets or higher numbers of trials?
- Clarity: Is the code well-organized and the report well-written?
- Creativity: Did the student add any additional features or unique approaches to the problem?

Submission:
Submit your project code and report in a single ZIP file.

Fill out the following list:
modul3_project week 6